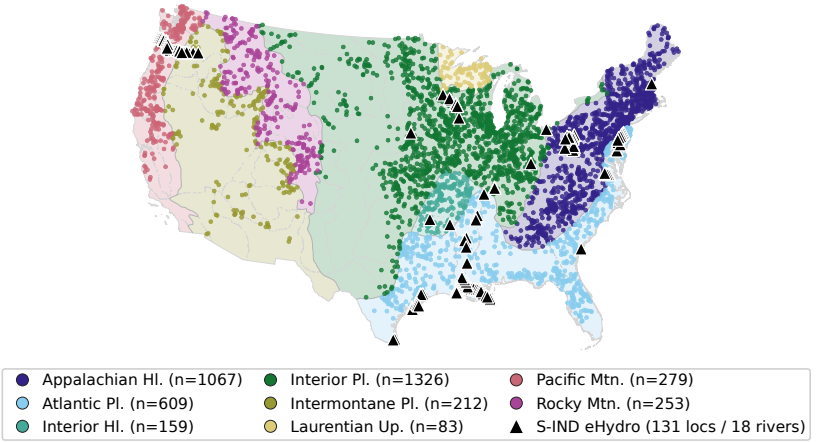
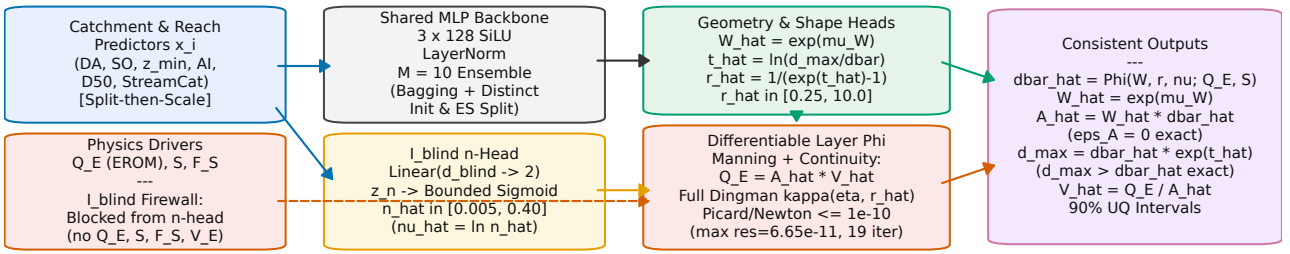
(a) CONUS Study Domain: n = 3988 HYDROSWOT Sites (8 F&J S-LRO Divisions) & S-IND [Descriptive]
S-IND: 131 locations / 18 rivers (EPSG:5070 Albers Equal Area)



(b) M5 Hard-Constrained Physics-Informed Architecture (I_blind + Differentiable Layer Phi) [Descriptive]
docs/3_experimental_protocol.md §4.1-§4.2; docs/2_mathematical_proofs.md §5-§7



(c) Validation Design (S-ID, S-LRO, S-TAIL, S-IND, S-RND) & Pre-Registered §4.7 Row 4 Outcome [Descriptive]

S-ID (3x5 Grouped CV; n = 3,988 sites)
Event 1 Audit: d_nn,med = 15.01 km
rho_hat = rho* = 169.23 km
Nested: strong=0.4213, weak=0.4879
Label: "optimistic under spatial/network dependence"

S-LRO (Leave-One-Region-Out OOD)
8 Held-Out F&J Divisions (n = 3,988)
Primary Spatial OOD Benchmark (H1, H3)
S-TAIL-DA: n = 678 upper-DA sites
S-TAIL-SO: n = 339 high-SO sites

S-IND (Independent USACE eHydro)
131 locations / 18 rivers
(131 locations / 18 rivers (UQ: 33 single-COMID rows))

S-RND [Secondary]: n = 9673 rows
(1/|I_j| weights across 3,988 sites)

Pre-Registered Decision Matrix Outcome (docs/3_experimental_protocol.md §4.7 row 4) [Confirmatory]:
Outcome: "M5 worse (S-LRO CI lower bound > 0)" | Venue: "ASCE; fallback per ADL-014"
row 4: "Diagnose with the pre-registered ablations (A1, A3, M4, SA1, SA5) and report. No post-hoc headline switch: any new variant is labelled exploratory"